Applications in Industry

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We recognise and pay respect to the Elders and communities – past, present, and emerging – of the lands that the University of Sydney's campuses stand on. For thousands of years they have shared and exchanged knowledges across innumerable generations for the benefit of all.

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Recap Week 9

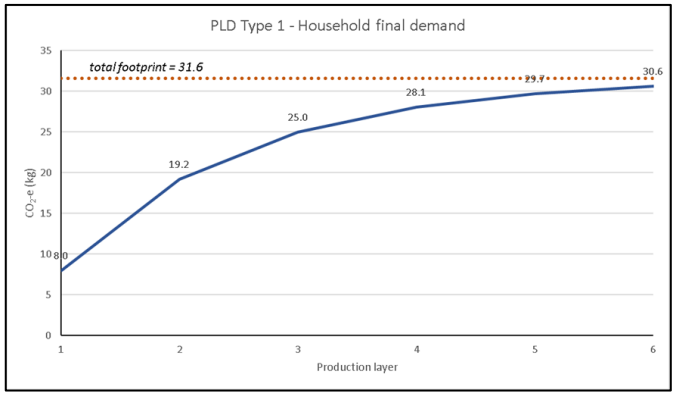
- The footprint equation can be decomposed using the Taylor series expansion

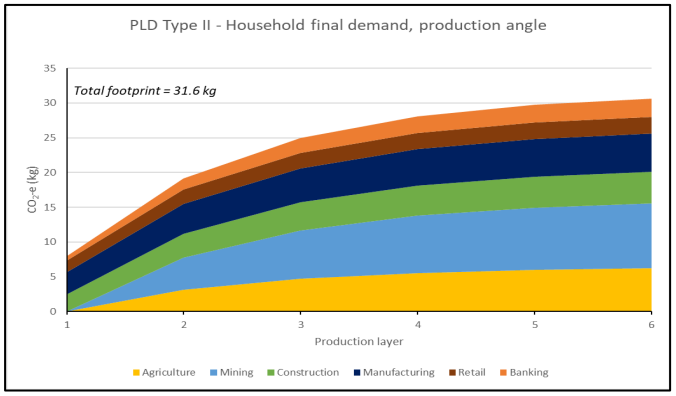
$$(\mathbf{I}-\mathbf{A})^{-1} = \mathbf{I} + \mathbf{A} + \mathbf{A}^2 + \mathbf{A}^3 + \mathbf{A}^4 + \cdots + \mathbf{A}^n$$

- This enables us to calculate the footprint at each production layer

- We can deconstruct this further by sector using transposition and element-wise multiplication

- Different types of charts to illustrate Production Layer Decomposition





Guest Lecture – Hannah-Jayne Shilling

From Academia to Industry

Final Assessments

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Homework for Week 10

Homework week 10 (Due: October 28, 2026)

The aim of this exercise is to apply all the techniques you have learnt to-date in a footprint analysis for a particular entity. You have been asked to calculate a modern slavery footprint assessment for a client, highlighting which of their consumption sectors has the greatest modern slavery footprint.

1. **Preparation:**

Create a modern slavery satellite matrix to use with your 14-sector input-output table. This new satellite matrix will be used for this week's homework. You can use hypothetical numbers for this satellite matrix, and do not need to construct it using the techniques used in week 5. Make sure that you include labels and units when you present the new satellite matrix.

2. **Calculate q and m for this indicator**

Use the formulae presented in week 6 to calculate q and m for the modern slavery indicator and your 14-sector input-output table.

3. **Construct an expenditure vector for the hypothetical client**

Create a new final demand vector for your client with data included against relevant sectors to reflect their annual expenditure.

4. **Calculate the total modern slavery footprint for this client's expenditure**

Use the equation qLy to calculate the total modern slavery footprint associated with this client's final demand, your modern slavery satellite matrix, and your input-output table.

5. **Calculate the modern slavery footprint for this client's expenditure by sector**

Diagonalise the client's final demand vector and use the new, diagonalised vector to calculate the modern slavery footprint for each sector.

6. **Determine the sector of expenditure that makes the greatest contribution to this client's total modern slavery footprint**

Which of the sector footprints is the highest?

7. **(Optional activity – not assessed) Provide your client with a Production Layer Decomposition chart based on their expenditure to provide a more detailed analysis of their modern slavery footprint.**

EXPLANATION: What is the difference between a consumption-based footprint and a production-based footprint? (100 words). Why would a consumption-based footprint be more useful than a production-based footprint for an Australian entity hoping to understand its own modern slavery footprint? (100 words)

Worked example for Week 10

Worked example of steps for an entity's carbon footprint analysis

Step 1 Obtain the entity's annual expenditure data, in the sectoral structure of your IO table*

* often the expenditure data is not provided in the correct structure, and a concordance is developed to allocate it to the appropriate sectors

	Entity expenditure AUD$
Agriculture	50
Mining	1
Construction	15
Manufacturing	341
Retail	250
Banking	850

Entity expenditure diagonalised					
50	0	0	0	0	0
0	1	0	0	0	0
0	0	15	0	0	0
0	0	0	341	0	0
0	0	0	0	250	0
0	0	0	0	0	850

Step 2 Obtain $\mathbf{q} = \mathbf{Q} * \hat{\mathbf{X}}^{-1}$ from the input-output table

	Agriculture	Mining	Construction	Manufacturing	Retail	Banking
CO2-e (kg) / $	0.00195	0.00660	0.00066	0.00066	0.00066	0.00066

Step 3 Obtain $\mathbf{m} = \mathbf{q} * \mathbf{L}$ from the input-output table

CO2-e (kg) / $	0.003156121	0.008745017	0.002667564	0.002648738	0.002871472	0.001587993

Step 4 Calculate the entity's footprint = $\mathbf{m} * \mathbf{y}_{ent}$

Homework Task 2

Submission requirements:

1. Submit your excel file as you did for the first homework submission, with one worksheet for each week. The new sheets will be weeks 6, 7, 8, 9 and 10.

2. Submit your answers to the 'EXPLANATION' questions for each week in a separate Word/PDF document. Ensure you note which week each explanation is referring to.

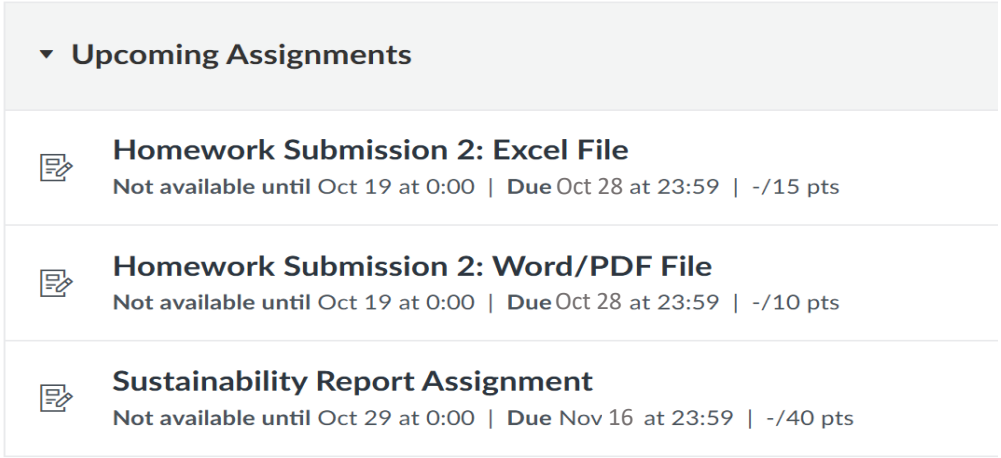
This assignment is due on October 28

Homework Task 2

Submission requirements:

3. Only upload one file to each of the submission links. The Explanation tasks should be submitted to the 'Word/PDF File' link, and the Excel file should be submitted to the 'Excel File' link .

4. If you load up more than one file to each submission, only the first file will be marked.



PHYS5033 organisation – assessment tasks

Tips for the Homework tasks:

- Make sure that you read the homework instructions thoroughly for each week – if you don't complete one of the steps expected, you won't get any marks for it.
- If there is a question posed in one of the Excel homework steps, make sure you provide an explicit answer for that question in the Excel file.
- Always include labels and units whenever you present results and make sure that your workings are included.
- You will be provided with worked examples and tutorial answers. These provide you with examples of how the calculations work – they do not provide you with a template for the homework. If you just replicate these, you will not be answering the steps expected in the homework.
- You should be able to answer the questions posed in the 'Explanation' part of the homework each week using the content provided in the lectures and in the background reading material. There is no need to use other sources to answer these questions, and doing so often results in students providing irrelevant or incorrect answers. If you do use other sources of information, these sources must be cited and referenced appropriately. You are strongly advised NOT to rely on AI tools to source your answers for these questions since they usually provide incorrect information.
- Make sure that your Excel submission is easy for the marker to read and find your results. This includes using appropriate font sizes, using appropriate row heights and column widths, and including text to note which step the results you're presenting relate to.

Council Sustainability Report

Weighting	Description
40%	**Sustainability report** This task requires students to prepare a sustainability report for a Council, providing quantitative figures for the calculation of their personal carbon footprint, based on techniques discussed in the lectures. Furthermore, students are required to provide a thorough discussion and a plan of action for reducing the footprint. Submissions are due on Turnitin by 11:59PM, 16 November, 2026. **See detailed guidelines below.**

think about the "audience"

Council Sustainability Report

Scenario

You have been asked by your local council to prepare a sustainability report outlining the methodology for calculating your carbon footprint as a resident of that council. Furthermore, the sustainability manager of the council has asked you to provide concrete examples and recommendations on ways for reducing your carbon footprint, and that of other residents of the council.

Note: This is an exercise to get you to engage with the technique of input-output analysis for calculation of individual footprints, and to get you thinking about ways of (and the need for!) reducing greenhouse gas emissions. If you do not wish to reveal your real spending patterns then you are free to use dummy data for the purpose of this assignment. It is important, however, to make sure that the dummy data that you use is realistic.

Council Sustainability Report

Part A – Quantitative calculations

- Collection of Expenditure Data: What do you spend money on? Collection of yearly expenditure data on various consumables (food products, electricity, fuel, transportation, clothing, furniture, services etc.).

- Construction of final demand shopping basket vector: Map your raw expenditure data to the 345 input-output categories.

- Calculation of CO_2-e footprint: Use the consumption-based footprint equation to calculate your carbon footprint.

- Comparison of per-capita footprints: Compare your carbon footprint figure with:
  a) per-capita footprint of an average Australian; and
  b) per-capita footprints in other countries.

  Present your results in the form of figures/tables. Example - see here: https://ourworldindata.org/co2-and-other-greenhouse-gas-emissions#co2-embedded-in-trade

Council Sustainability Report

Part B – Report Writing

Put together a report for your local council showcasing the results of the assessment, and outlining strategies for reducing carbon emissions. Your report should include the following information:

- *Introduction*: The need for a reduction in greenhouse gas emissions. Provide a national and a global perspective;
- *Introduction*: The case for the use of quantitative methods, such as input-output analysis, for analysing footprints;
- *Methodology*: The methodology underlying input-output analysis;
- *Methodology*: Limitations of the input-output technique. (See Canvas for ideas here)

Council Sustainability Report

Part B – Report Writing

- Results: Present your results from the input-output assessment. You should include (but are not limited to) information on your carbon footprint, and a comparison of your carbon footprint with average per-capita carbon footprints in Australia and other countries. It is expected that students will engage with the techniques discussed in the unit to present results at a sector-level.

- Discussion: Provide a discussion of the results in the broader context of sustainability. Why do this assessment?

- Recommendations: Provide concrete recommendations to the sustainability manager of your local council for steps that you will take for reducing your carbon footprint, and the steps the council can take for facilitating/assisting/helping you in reducing your footprint and that of other individuals in the area. Recommendations should relate to your results.

Council Sustainability Report

Part B – Report Writing

- Supplementary Information: Include a supplementary information section with evidence of the process you followed for undertaking the calculations - this could either be in the form of Excel/ MATLAB snapshots. The snapshots should show the process you followed for carrying out the calculations, and should be labelled correctly.

Please submit **one** file, either as Word or PDF. Please include snapshots from your Excel/MATLAB calculations in the same file, as supplementary information. Do not submit more than **one** file.

This assignment is due on November 16, 2026
Word limit is 3,000 words

Notes regarding submissions

Assignments (and other assessment items) submitted late without permission will incur a late penalty equal to 5% of the maximum mark per day. These deductions will continue for 10 calendar days or until a solution for the assignment is released or marked assignments are returned to other students. At that point the available mark will go to zero. For example, on an assignment given a mark of 7/10, the penalty would be 0.5 marks if submitted up to 24 hours late, resulting in a final mark of 6.5/10. If the assignment is submitted 6 days late, the penalty would be 3 marks and the final mark would be 4/10.

Assignments that are **over the word limit** as stipulated in the assessment guidelines will incur a 5% per 100 words, capped at 40%. For example, for the Sustainability Report, this would be:

3,000 – 3,300	0 penalty
3,301 – 3,400	2 mark penalty
3,401 – 3,500	4 mark penalty
3,501 – 3,600	6 mark penalty
3,601 – 3,700	8 mark penalty
3,701 – 3,800	10 mark penalty

If an incorrect file is submitted for an assignment and a correct file is provided after the due date, a late penalty will still be applied.

Council Sustainability Report

You have been provided with the following data files for Year 2018:

Intermediate demand matrix (T) – 345 x 345 matrix (data on 345 sectors of the Australian economy. See 'Labels file' for more information). The units for these data are Australian million dollars. For example, a value of 2 in a cell means 2 million Australian dollars.

Total output vector (x) – 345 x 1 vector (data on total output for 345 sectors of the Australian economy). The units for these data are Australian million dollars.

Satellite account (Q) – 1 x 345 vector (data on carbon dioxide equivalent emissions for 345 sectors of the Australian economy). The units for these data are kilotonnes.

These files are stored in two directories: MATLAB files and Excel files. Data in both directories are the same. You can complete this assignment using either the MATLAB programming software (in that case use the MATLAB files) or MS Excel (use Excel files).

Council Sustainability Report

The marking rubric is there to help you!

ASSESSMENT MATRIX – SUSTAINABILITY REPORT		
Total Mark	**out of 40**	
Category	**Criteria**	**Total marks**
Presentation (10 marks)	No grammatical, typographical and spelling errors.	3
	All three elements are addressed. • Information is presented in a coherent and logical sequence. • Legible and well set-out. Use of single/double spacing is up to the student to decide. • The text flows well from one paragraph to next.	3
	Quality of presentation – the assignment is written in professional academic English	2
	Well-structured with appropriate use of headings/sub-headings and figures/tables.	2

Council Sustainability Report

Content (27 marks)	Presents a comprehensive introduction on the need for reducing greenhouse gas emissions; including both a national and a global perspective; and the need for quantitative methods	5
	Adequately explains the technique of input-output analysis for footprint assessments, ensuring the information is provided in a format that is readily accessible for a non-expert reader (- in this case sustainability manager of a council).	4
	Input-output equations are adequately presented in the report	3
	Comprehensively presents the results of the footprint assessment, including, but not limited to, a comparison of per-capita footprints with those of an average Australian and for other countries as well.	4
	Provides an excellent discussion of the results in the broader context of sustainability.	4
	The conclusions/recommendations of the report are comprehensive, feasible and reasonable. Recommendations are provided both at an individual and council level.	4
	Supplementary Information document is provided, which includes evidence of the process the student followed for carrying out the carbon footprint assessment	3

Council Sustainability Report

The marking rubric is there to help you!

References (3 marks)	• References listed within the report • References listed at the end of the report. • References cited/formatted correctly. Any reference format is OK. However, students should pick ONE format and be consistent with it.	3
TOTAL		**40**